# Dividing Fractions

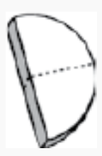
Division and multiplication are opposite operations. This important fact allows you to change division into multiplication.

To divide a fraction, change the division symbol to multiplication and invert the number you are dividing by. The inverted fraction is the **reciprocal** of the original fraction.

## Example 1

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Angela has to give her cat $\frac{1}{4}$ of a pill twice a day. If she has $\frac{1}{2}$ of a pill left, how many more times can she give her cat its medicine?



**Step 1**

Invert the divisor (the number you are dividing *by*).

$\frac{1}{2} \div \frac{1}{4}$ $\frac{1}{4}$ reciprocal $\frac{4}{1}$ inverts to

**Step 2**

Change ÷ into ×. Use canceling, if possible.

$\frac{1}{1\cancel{2}} \times \frac{\cancel{4}^{2}}{1} \rightarrow \frac{1}{1\cancel{2}} \times \frac{\cancel{4}^{2}}{1} = \frac{2}{1} = 2$

**Step 3**

Multiply across using the new numerators and denominators.

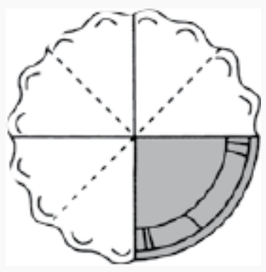
$\frac{1}{1} \times \frac{2}{1} = 2$

Angela can give her cat $\frac{1}{4}$ of a pill **2 more times.**

## Example 2

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The Morrison family has $\frac{3}{4}$ of a pie left over. If there are 6 people in the family, what fraction of the original pie will each person get?



**Step 1**

Write the whole number as a fraction with a denominator of 1. Then invert the fraction.

$\frac{3}{4} \div \frac{6}{1}$ $\frac{6}{1}$ reciprocal $\frac{1}{6}$ inverts to

**Step 2**

Change ÷ into ×. Use canceling, if possible.

$\frac{1\cancel{3}}{4} \times \frac{1}{\cancel{6}_{2}} \rightarrow \frac{1\cancel{3}}{4} \times \frac{1}{\cancel{6}^{2}} = \frac{1}{8}$

**Step 3**

Multiply across using the new numerators and denominators.

$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

Each person will get $\frac{1}{8}$ of the pie.

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Lesson 5: Operations with Fractions

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